

Section II: Case III

**Defined Contribution Plan and MPF Retirement
Planning**

AMA1D03 Introduction to Pension Mathematics

(a) DC Plan and MPF in Hong Kong's Pension System

1. DC versus DB

In a **defined contribution (DC)** plan, contributions are fixed but the retirement benefit is not. The final balance depends on contributions, investment returns, and fees, and the member bears most investment and longevity risk. In a **defined benefit (DB)** plan, the benefit formula is fixed while required contributions are not; the sponsor bears most funding risk. Hong Kong's MPF, launched **1 December 2000**, is a mandatory DC scheme and forms the **second pillar**, together with ORSO where applicable.

Hong Kong has no public pension linked to past earnings. The **first pillar** provides flat-rate, means-tested cash benefits: CSSA as the last-resort safety net, OALA for lower-income seniors, and OAA which is more widely available but pays less. The **third pillar** is personal savings, insurance, and family support. For the middle class, MPF is the main retirement base.

2. MPF Rules Relevant to Case III

The MPFA classifies schemes into master trust, employer-sponsored, and industry schemes. Each member holds a portable personal account.

Mandatory contributions: employee and employer each pay **5%** of relevant income, with a monthly floor of HK\$7,100 and ceiling of HK\$30,000. If income is below HK\$7,100, the employee pays nothing but the employer still contributes 5% of HK\$7,100. The maximum from either party is HK\$1,500 per month.

Voluntary contributions: **In-scheme VC** has no salaries-tax relief and follows mandatory withdrawal rules. **Tax-deductible voluntary contributions (TVC)** are held in a separate account; up to **HK\$60,000 per year** is deductible for salaries tax.

Members can normally withdraw at age **65**, or earlier on permanent departure, incapacity, or terminal illness. They may take a lump sum or convert part of the balance into an annuity to manage longevity risk. Fund fees are typically 0.5%–2% per year. In Case III, Peter and Mary each contribute HK\$1,000 with employer matching, totalling **HK\$2,000** per month. Peter invests in a global equity fund and Mary in a Hong Kong equity fund. Mandatory contributions alone are **not enough** to maintain pre-retirement living standards.

(b) Analysis: Peter and Mary

1. Modelling Setup

We simulate monthly MPF accumulation from December 2000 until each member reaches age 65. Table 1 lists the main assumptions and data sources.

Notes: Salary growth linked to CPI implies flat real wages; actual career wage growth would lower the replacement ratio further. Returns use historical data to the latest available month, then **4% p.a.** nominal forward returns. The **4% withdrawal rule** sets first-year retirement income at 4% of the balance. We assume a **1% p.a.** expense ratio, near the middle of the MPF range.

Table 1: Modelling assumptions and data sources

Item	Value	Source / note
Initial monthly salary	HK\$20,000	Case III
Monthly MC, employee plus employer	HK\$2,000	5% + 5%
Salary growth	HK Composite CPI	C&SD annual YoY
Mary: Hong Kong fund	Hang Seng Index $\hat{HSI}$	Yahoo Finance
Peter: global fund	Spliced $\hat{GSPC}$, EFA, ACWI	Yahoo Finance
Forward return / fee	4% p.a. / 1% p.a.	After latest data
Retirement income	4% withdrawal rule	Year-1 income = 4% of balance

Fund proxies. Mary is assigned a Hong Kong equity fund. We use the **Hang Seng Index** $\hat{HSI}$ as its return proxy because the Tracker Fund of Hong Kong (2800.HK) replicates this index and Yahoo Finance provides monthly returns from December 2000.

Peter is assigned a global equity fund. No single global ETF covers the full MPF period, so we splice three series: $\hat{GSPC}$ from Dec 2000 to Aug 2001; 65% $\hat{GSPC}$ plus 35% EFA from Sep 2001 to Mar 2008; ACWI from Apr 2008 onward.

Table 2: Case III participants

	Peter	Mary
Start age, Dec 2000	20	40
Retirement age	65	65
Contribution years	45	25
Retirement year	2045	2025
Assigned fund	Global equity fund	Hong Kong equity fund
Return proxy	Spliced global index	Hang Seng Index

2. Baseline Results

Table 3: Baseline simulation results

	Peter	Mary
Contribution years	45.1	25.1
Total contributions, HK\$	1,443,931	723,931
Investment gains, HK\$	3,552,431	126,930
Final MPF balance, HK\$	4,996,362	850,862
Final annual salary, HK\$	555,156	374,107
Monthly MPF income, HK\$	16,655	2,836
Replacement ratio	36.0%	9.1%

Peter's balance rises steadily over 45 years. Mary's balance over 25 years ends at about one-sixth of Peter's level. Peter's investment gains of HK\$3.55 million exceed his contributions of HK\$1.44 million, illustrating long-horizon compounding. Hong Kong equities were volatile over Mary's 2000–2025 period, which also helps explain her lower balance. Figure 1 separates **historical fund**

returns from projected returns: Mary's entire path uses actual index data through retirement in 2025; Peter uses historical data through June 2026, then a 4% p.a. forward assumption until 2045.

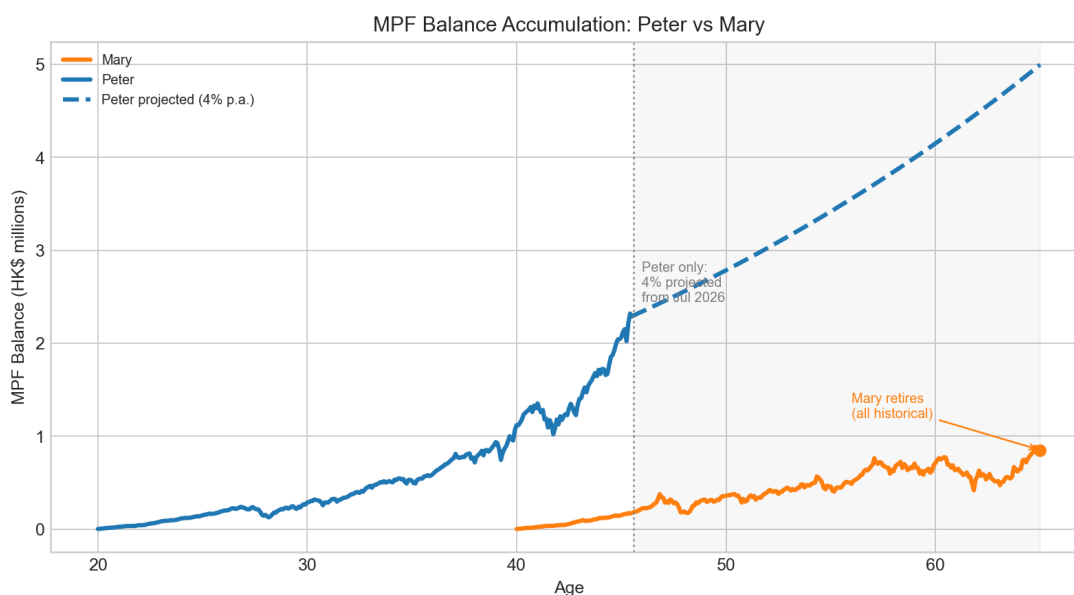


Figure 1: MPF balance by age. Mary: solid line only, all historical index returns through retirement in Dec 2025. Peter: solid line uses historical returns through Jun 2026; dashed line is Peter only, projected at 4% p.a. to 2045.

3. Advantages and Disadvantages

Advantages:

1. Individual ownership and portability suit Hong Kong's mobile labour market.
2. Employer matching doubles savings without extra employee cost.
3. Members can choose funds aligned with risk tolerance.
4. TVC of up to HK\$60,000 per year is tax-deductible.
5. Early starters benefit most from compounding; at 4% p.a., wealth roughly doubles every 18 years.

Disadvantages:

1. Members bear full market and sequence risk.
2. Unlike DB plans, there is no pooling of longevity or investment shortfalls.
3. Even Peter reaches only 36% replacement on mandatory contributions.
4. Fees compound against members over 25–45 years.
5. Mary's 9.1% ratio shows that starting at 40 cannot be fixed by fund choice alone.

Near retirement, members may adjust fund allocation: Peter can shift from global equity toward bonds and capital-preservation funds after age 55; Mary may move from Hong Kong equity toward mixed-asset funds to reduce volatility. Part (c) builds on these balances and examines how MPF can be combined with OALA.

4. Start Age versus Fund Choice

Table 4 reports replacement ratios by start age. Table 5 shows fund-swap sensitivity. Figure 2 plots the results.

Table 4: Replacement ratio by start age and fund

Start Age	Fund	Final Balance, HK\$	RR
20	Global	4,996,362	36.0%
20	Hong Kong	2,427,886	17.5%
30	Global	3,417,078	30.0%
30	Hong Kong	1,500,213	13.2%
40	Global	2,063,587	22.1%
40	Hong Kong	850,862	9.1%
50	Global	520,922	6.9%
50	Hong Kong	457,617	6.0%

Table 5: Fund-swap sensitivity

Name	Assigned RR	Swapped RR	Change, pp
Peter, age 20	36.0%	17.5%	−18.5
Mary, age 40	9.1%	22.1%	+13.0

Key finding: At the same fund, starting at 20 rather than 40 widens the replacement ratio by 27 percentage points. Peter at 20 with the weaker Hong Kong fund still ends above Mary. **Contribution horizon dominates fund selection.**

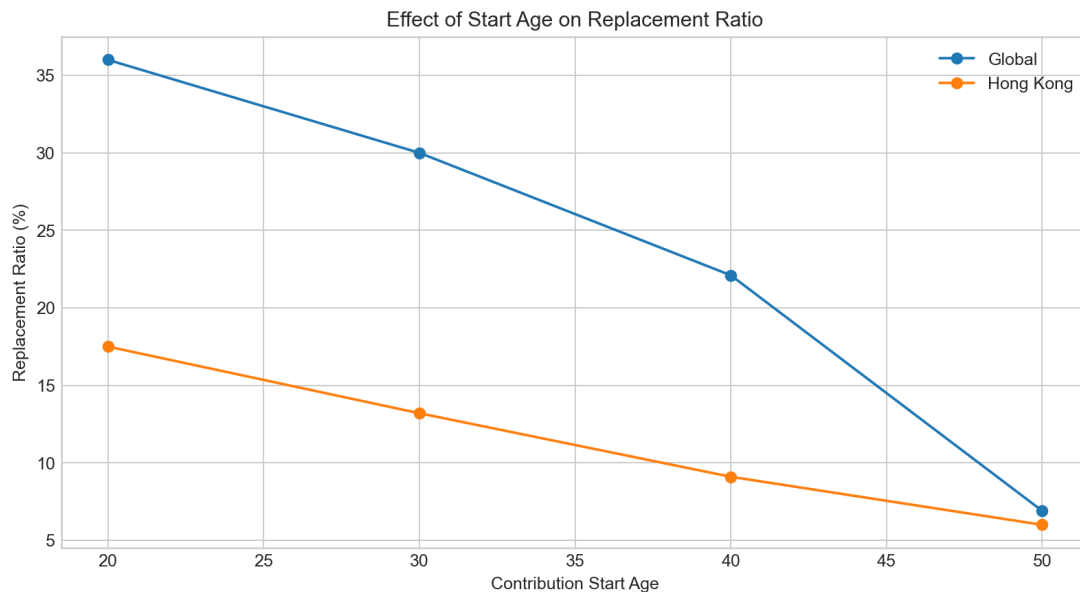


Figure 2: Effect of start age on replacement ratio

5. Gaps and TVC

Planners often target **60–70%** replacement. Using final monthly salary from Table 3, the gap to 60% is about HK\$11,103 per month for Peter and HK\$15,870 for Mary.

Table 6: Extra monthly TVC to reach target replacement ratio

Member	Current RR	Target RR	Extra TVC, HK\$
Peter	36.0%	60%	1,582
Peter	36.0%	70%	2,242
Mary	9.1%	60%	13,205
Mary	9.1%	70%	15,799

Peter needs about HK\$1,600 per month of **TVC** (tax-deductible voluntary contributions) to reach 60%, within the annual limit of HK\$60,000. Mary needs HK\$13,205 per month, far above that cap; only the first HK\$5,000 per month receives tax relief.

(c) Social Security and Recommendations

1. Three Pillars and OALA

Part (c) asks how DC retirees combine MPF with first-pillar OALA. OALA pays Normal allowance of HK\$2,065 or Higher allowance of HK\$4,185 per month in 2025/26, subject to income and asset tests. For a single person, assessable assets are capped at about HK\$401,000 and monthly assessable income at about HK\$10,700 for Higher OALA.

Unwithdrawn MPF counts as assets. To use OALA alongside MPF, retirees may need to **structure benefits at age 65**. We assume **Mary annuitises her MPF** through the MPF platform so benefits are taken as a pension stream rather than a retained lump sum, keeping assessable liquid assets below the cap. Her MPF pension of HK\$2,836 per month is below the income limit, so she qualifies for **Higher OALA of HK\$4,185**. **Peter** has a balance of HK\$4.99 million and MPF drawdown of HK\$16,655 per month, so he fails both the asset and income tests and receives no OALA.

2. Combined Retirement Income

Table 7 reports the integrated Part (c) plan: MPF pension income plus OALA where eligible. Figure 3 compares the two members.

Table 7: MPF plus OALA retirement income (Part c)

Member	MPF income	OALA	Total	Combined RR
Peter	16,655	0	16,655	36.0%
Mary	2,836	4,185	7,021	22.5%

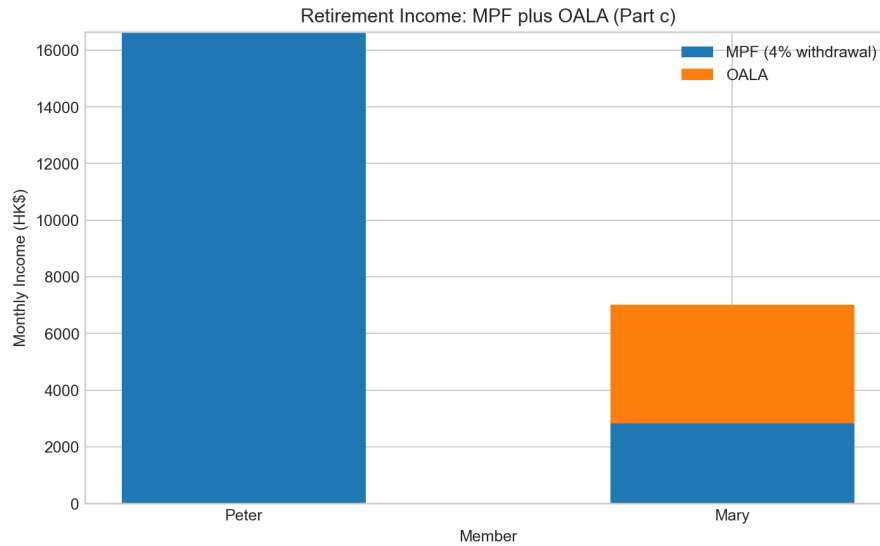


Figure 3: Retirement income: MPF plus OALA

3. Strategies and Conclusions

Peter, retiring in 2045:

- MPF provides a 36% base but OALA is unavailable.
- Add about HK\$1,600/month through TVC to reach 60% within the tax-deductible limit.
- Shift gradually toward bonds and capital-preservation funds after age 55 to reduce sequence risk.
- Consider partial annuitisation for longevity protection.

Mary, retiring in 2025:

- **Annuitise MPF at 65** and apply for **Higher OALA** to raise combined income to HK\$7,021 per month, 22.5% replacement.
- OALA helps but the gap to 60% remains large; add TVC where affordable and third-pillar savings or part-time work.
- Delaying retirement to 70 raises MPF-only RR only to 10.9% (Table 8).

Table 8: Mary: delayed retirement sensitivity

Retire age	Years	Balance, HK\$	RR
65	25.1	850,862	9.1%
67	27.1	930,635	9.6%
70	30.1	1,128,770	10.9%

A mixed plan for Mary of MPF plus Higher OALA plus about HK\$5,000 per month from other sources gives roughly HK\$12,021 per month, still below the 60% target of HK\$18,706.

Table 9: Summary recommendations

	Peter	Mary
Mandatory MPF sufficient?	No, 36%	No, 9.1%
Add TVC?	Yes, moderate	Yes, urgent
Social security fills gap?	No, asset test	Partially, OALA

Core conclusions:

1. Mandatory MPF alone cannot maintain pre-retirement living standards.
2. **When contributions begin** matters more than fund choice.
3. OALA can supplement MPF for members like Mary who structure benefits to meet the tests; high-balance members like Peter rely on MPF and voluntary savings instead.
4. A complete plan combines MPF, TVC within tax limits, third-pillar savings, lifecycle de-risking, and partial annuitisation.

Key Formulae

Future value with CPI-linked contributions:

$$FV = \sum_{t=1}^n PMT_0(1+g)^{t-1} \prod_{k=t+1}^n (1+r_k) \quad (1)$$

When r and g are constant and $r \neq g$,

$$FV = PMT_0(1+r) \cdot \frac{(1+r)^n - (1+g)^n}{r-g} \quad (2)$$

Replacement ratio and monthly gap to a 60% target:

$$RR = \frac{FV \times 0.04}{Salary_{final}}, \quad Gap_{monthly} = 0.60 \times \frac{Salary_{final}}{12} - \frac{FV \times 0.04}{12} \quad (3)$$

where PMT_0 is initial monthly contribution, g is contribution growth, r_k is net return in period k , and n is the number of periods.

References

- [1] MPFA, *MPF Fund Performance Platform*, <https://mfp.mpfa.org.hk/>.
- [2] SWD, *Old Age Living Allowance*, <https://www.swd.gov.hk/oala/>.
- [3] C&SD, *Composite Consumer Price Index*.